Sample Test Plan
(for Program 1 from last week's Programming Assignment 3)

Test Conditions
(Minimal conditions that must be tested in order to validate program correctness)

1. Ounces input
   a. Under 16 ounces
   b. Between 16 and 31 ounces
   c. 32 ounces or more

2. Container type input (reads, displays with a word, and charges correctly)
   a. cup
   b. cone
   c. Accepts and correctly interprets lowercase entry
   d. Accepts and correctly interprets uppercase entry

3. Toppings wanted (reads and discounts correctly)
   a. wants toppings
   b. does not want toppings
   c. Accepts and correctly interprets lowercase entry
   d. Accepts and correctly interprets uppercase entry

4. Number of toppings (only prompts if 3 is wants toppings)
   a. 1
   b. more than 1

5. 5 punches on card (reads and charges correctly)
   a. no
   b. yes

Test cases for program, and which condition will be tested:

Test 1:	Will test conditions 1a, 2a, 2c, 3b, 3d, 5a
Test 2:	Will test conditions 1b, 2b, 2c, 3a, 3d, 4a, 5b
Test 3:	Will test conditions 1b, 2a, 2d, 3a, 3c, 4b, 5b
Test 4:	Will test conditions 1c, 2b, 2d, 3b, 3c, 5a

Note that between all 4 test cases, ***all*** *test conditions will be tested.*
(If all were not covered, more test cases would be needed)

See actual tests on next page.

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Program 1 Test Data	**Test 1** Conditions 1a, 2a, 2c, 3b, 3d, 5a	**Test 2** Conditions 1b, 2b, 2c, 3a, 3d, 4a, 5b	**Test 3** Conditions 1b, 2a, 2d, 3a, 3c, 4b, 5b	**Test 4** Conditions 1c, 2b, 2d, 3b, 3c, 5a
TEST INPUT				
Ounces	15	16	31	32
Cup or Cone	p	w	P	W
Toppings wanted	N	T	t	n
Num Toppings		1	5	
5 punches	no	yes	yes	no
EXPECTED OUTPUT				
Weight	15	16	31	32
Container	plastic cup	waffle cone	plastic cup	waffle cone
Num Toppings	0	1	5	0
Yogurt Cost	6.60	7.04	13.64	14.08
Plus Cone		0.57		0.57
Plus Toppings	0.00	0.63	3.15	0.00
Subtotal		8.24	16.79	14.65
Less Discount		0.72	0.72	1.58
Bill Total		7.52	16.07	13.07
Reduced	6.60	3.76	8.04	

Note that the number of ounces chosen to use as specific test inputs (15, 16, 31, and 32) were chosen because they each represent border cases for the discount ranges.

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